Vade Retro Coriandre: Enzymatic Bioconversion and Supramolecular Taste-Masking of (E)-2-Decenal for Coriander-Sensitive Individuals

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Abstract — Coriander (*Coriandrum sativum* L.) is a highly divisive culinary herb due to the presence of specific unsaturated aldehydes, primarily *(E)*-2-decenal, which trigger an unpleasant soapy taste in individuals possessing the OR6A2 olfactory receptor gene. This study investigates two distinct remediation strategies: enzymatic bioconversion of the volatile aldehyde using baker's yeast (*Saccharomyces cerevisiae*) and supramolecular taste-masking via encapsulation in β-cyclodextrin (β-CD). While steam distillation of fresh coriander leaves yielded negligible essential oil due to scale limitations and low natural concentrations ($< 0.3\%$), biochemical reduction was attempted using commercial *(E)*-2-decenal. Infrared spectroscopy of the bioconversion product revealed an incomplete reaction, characterized by the persistence of the carbonyl ($C = O$) stretch at 1709 cm^{-1} and the absence of an alcohol ($O - H$) band. In contrast, supramolecular encapsulation via co-precipitation in aqueous ethanol at 55°C proved successful. ^1H NMR spectroscopy validated inclusion complex formation through diagnostic upfield shifts of the internal cavity protons. Furthermore, headspace SPME/GC-MS demonstrated a reduction in volatile release. Finally, a green chemistry assessment revealed that the cyclodextrin masking approach is both more ecologically sustainable (0.101 kg CO_2e vs. 0.303 kg CO_2e) and cost-effective (EUR 6.17 vs. EUR 8.07) than the bioconversion process.

1. INTRODUCTION

Coriander (*Coriandrum sativum* L.) is an herb utilized globally, yet it remains one of the most polarizing culinary ingredients [1]. For a significant portion of the population, genetic variations in the olfactory receptor gene *OR6A2* cause the herb's aroma to be perceived as soapy and bitter [1]. Analytical characterization of coriander leaves indicates that the characteristic aroma is governed by a series of volatile aliphatic aldehydes, notably *(E)*-2-decenal, *(E)*-4-decenal, and *(E)*-2-dodecenal [2]. Among these, *(E)*-2-decenal is the primary contributor to the aroma, constituting up to 32% of the volatile leaf profile [2]. Because these unsaturated aldehydes share highly similar chemical reactivities, the present study focuses exclusively on *(E)*-2-decenal. Nevertheless, the remediation strategies proposed here can be transposed to all other related derivatives.

To address this issue, the project "Vade Retro Coriandre" explores two chemical strategies to remediate the dish directly or block sensory perception in the mouth. Both methods must prioritize non-toxic, food-safe, and biocompatible agents. The first approach acts directly on the food matrix by converting the aldehydes into less odoriferous alcohols using enzymes present in baker's yeast (*Saccharomyces cerevisiae*). The second approach targets the sensory receptors by encapsulating the free aldehyde in the mouth using supramolecular hosts, specifically cyclodextrins, to suppress volatility and taste perception.

2. MATERIALS AND METHODS

2.1 Detection and Identification of (E)-2-Decenal from fresh coriander leaves

Headspace Solid-Phase Microextraction coupled with Gas Chromatography-Mass Spectrometry (HS-SPME/GC-MS) was employed to detect the volatile aldehydes from fresh mashed coriander leaves. The samples were heated at 70°C in sealed vials, and volatiles were extracted using a DVB/CAR/PDMS SPME fiber before thermal desorption in the GC injector port. The mass spectra were subsequently extracted and analyzed for the detection of *(E)*-2-decenal.

2.2 Distillation and Extraction Challenges

An initial attempt was made to isolate natural essential oils from fresh coriander leaves. A mixture containing 57.24 g of finely minced coriander leaves, 3.0 g of sodium chloride, and water was subjected to steam distillation. NaCl was added to increase the osmotic pressure, promoting cell lysis and volatile release. The mixture was heated at 150°C for 4 hours. Due to scale and connector limitations, a larger round-bottom flask could not be integrated, and no observable organic phase was recovered. Consequently, commercial *(E)*-2-decenal ($> 95\%$, Sigma-Aldrich) was utilized for all subsequent bioconversion and encapsulation trials.

2.3 Yeast-Catalyzed Bioconversion

Bioconversion of *(E)*-2-decenal was performed adapting protocols for green-note aldehyde reduction [3]. Active dry yeast (120 g/L) was hydrated in warm water (30–35°C) for 10 min, followed by the addition of sucrose (30 g/L) to stimulate fermentation and regenerate intracellular NADH cofactors. After 45 min of prefermentation, a solution containing 1.4 mL of commercial *(E)*-2-decenal dissolved in 14 mL of absolute ethanol (acting as a biocompatible co-solvent to enhance solubility) was added dropwise.

The reaction was stirred at 30°C for 48 hours. Following incubation, Celite was added as a filter aid, and the yeast suspension was filtered. The aqueous filtrate was extracted twice with 50 mL of diethyl ether. The combined organic phases were dried over anhydrous magnesium sulfate ($MgSO_4$), filtered, and concentrated under reduced pressure to yield a crude oil for FTIR and NMR analysis.

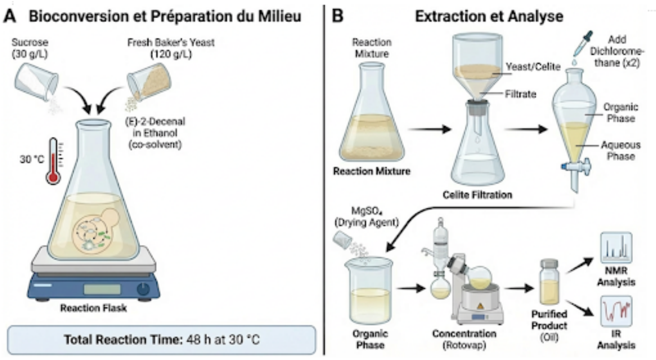


Figure 1: Experimental setup and downstream extraction protocol for the whole-cell bioconversion using *S. cerevisiae.*

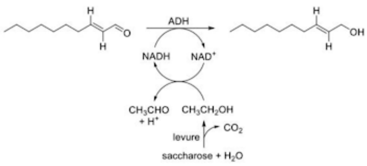


Figure 2: Reaction pathway for the enzymatic reduction of *(E)*-2-decenal to the corresponding alcohol via yeast alcohol dehydrogenase (ADH) [3].

2.4 Cyclodextrin Taste-Masking

Host-guest complexation was investigated using β-cyclodextrin (β-CD) as the supramolecular carrier. Assuming that the linear C10 chain of *(E)*-2-decenal has similar chemical affinities and dimensional constraints as a C10 carboxylic acid, β-CD (0.78 nm internal cavity diameter) was selected over γ-CD (0.95 nm). Hydrophobic van der Waals interactions between *(E)*-2-decenal and the complexing agent are thus maximized.

As demonstrated in Table 1, it can be observed that the association constant (K_{CDA}) is significantly higher for the complexation of the C10 aliphatic chain with β-cyclodextrin than with the wider γ-cyclodextrin [5].

Table 1: Association constants (K_{CDA}, M^{-1}) for β- and γ-cyclodextrins with C8 and C10 carboxylic acids [5].

System	$K_{CDA} = [\mathbf{CDA}]/([\mathbf{CD}][\mathbf{A}])$
βCD-C_{10}	$(2.6 \pm 0.2) \times 10^3$
γCD-C_{10}	$(2.5 \pm 0.5) \times 10^2$
βCD-C_8	$(5.1 \pm 0.2) \times 10^2$
γCD-C_8	$(4.7 \pm 0.2) \times 10^1$

The inclusion complex was synthesized using a co-precipitation method. β-CD (10 g) was dissolved in 100 mL of a 1:2 ethanol/water mixture heated to 55°C. Concurrently, 2 mL of *(E)*-2-decenal was dissolved in 20 mL of ethanol and added dropwise under vigorous stirring. The mixture was maintained at 55°C for 4 hours, then cooled slowly and stored at 4°C overnight. The resulting white crystalline precipitate was isolated by filtration, washed with cold ethanol, and dried to obtain a fine powder.

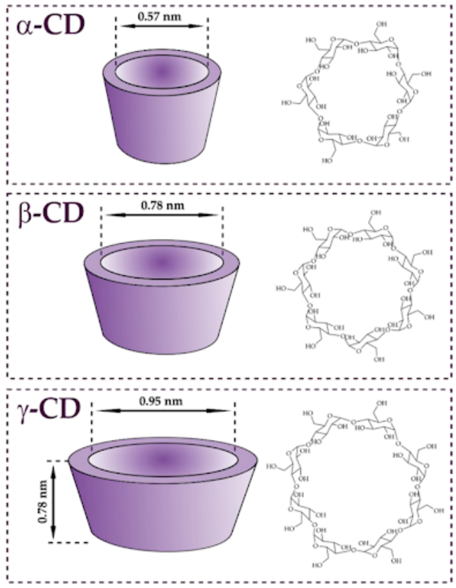


Figure 3: Structural dimensions and internal cavity diameters of α-, β-, and γ-cyclodextrins [6].

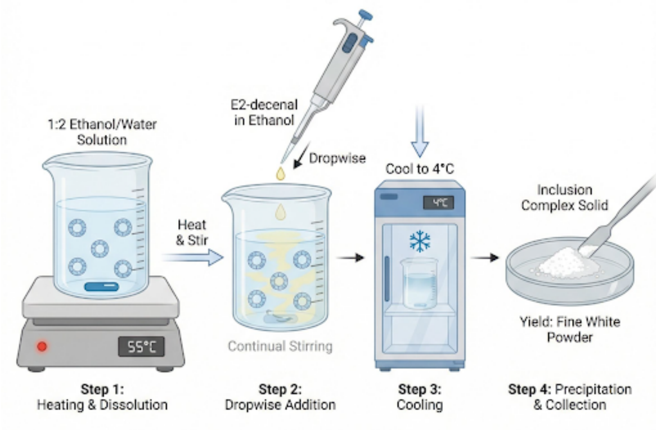


Figure 4: Experimental protocol for the supramolecular encapsulation of *(E)*-2-decenal into β-cyclodextrin via co-precipitation.

2.5 Analytical Techniques

Headspace Solid-Phase Microextraction coupled with Gas Chromatography-Mass Spectrometry (HS-SPME/GC-MS) was employed to evaluate the volatile profiles. ^1H NMR spectra were recorded in D_2O or $CDCl_3$ to observe chemical shift deviations. FTIR spectra were acquired on a Bruker spectrometer to monitor functional group transformations.

3. RESULTS AND DISCUSSION

3.1 Detection and Identification of (E)-2-Decenal from fresh coriander leaves

To confirm the presence of *(E)*-2-decenal as a primary volatile compound in fresh coriander, headspace solid-phase microextraction coupled with gas chromatography-mass spectrometry (HS-SPME/GC-MS) was performed. The resulting chromatogram (Figure 5) exhibited a prominent peak at a retention time of approximately 24.5 min, which was tentatively assigned to *(E)*-2-decenal based on its high relative abundance.

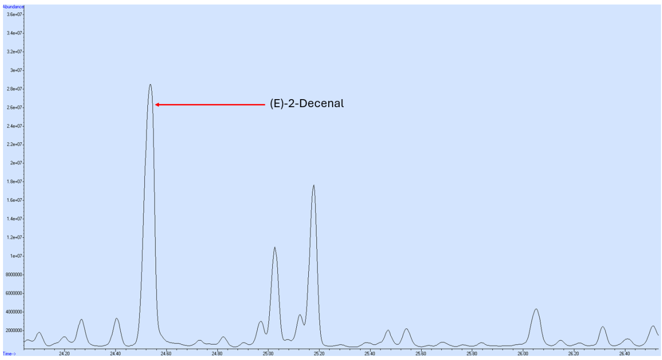


Figure 5: GC-MS chromatogram of the volatile profile from fresh coriander leaves, highlighting the major peak corresponding to *(E)*-2-decenal.

The mass spectrum of this specific peak was extracted (Figure 6 and compared against the NIST standard reference database (Figure 7). The experimental spectrum (red, top) perfectly matches the library spectrum for *(E)*-2-decenal (blue, bottom). Both the presence of the molecular ion and the relative intensities of the major fragments (m/z 70, 55, 41, 83, 98, 110) align, confirming that *(E)*-2-decenal is a highly abundant constituent of the coriander leaf headspace.

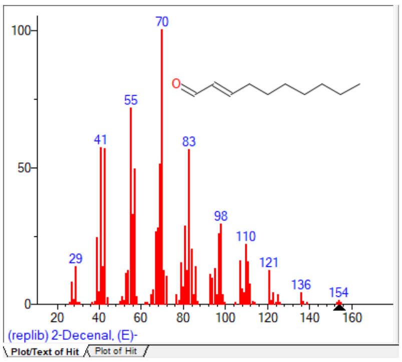


Figure 6: Obtained mass spectrum of the largest peak, identified as *(E)*-2-decenal after comparison.

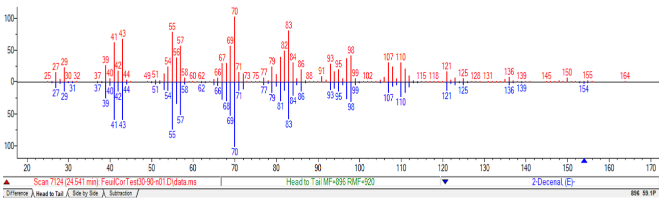


Figure 7: Head-to-tail comparison of the experimental mass spectrum (top, red) and the NIST database reference spectrum for *(E)*-2-decenal (bottom, blue).

3.2 Distillation Inefficiency Analysis

The failure of steam distillation to yield an observable essential oil phase is attributed to two factors: the low overall concentration of essential oil in fresh coriander leaves (< 0.3 wt%) and the highly diluted headspace of the system. In literature, steam distillation typically requires 100–500 g of fresh foliage to yield 50–200 μL of oil. Because our scale was restricted to 57.24 g of biomass, the mass of extracted volatiles was below the holdup volume of the glass connectors. Additionally, *(E)*-2-decenal has a moderate boiling point of 70°C at reduced pressure, meaning standard simple distillation or direct solvent extraction would be highly superior to co-distillation which overly dilutes the target compounds in water.

3.3 Bioconversion Performance

In yeast, alcohol dehydrogenases (ADH) catalyze the reduction of aldehydes via a zinc-coordinated active site [4]. The carbonyl oxygen coordinates to a Zn^{2+} ion, polarizing the double bond and facilitating hydride transfer from NADH, followed by protonation to yield the corresponding fatty alcohol.

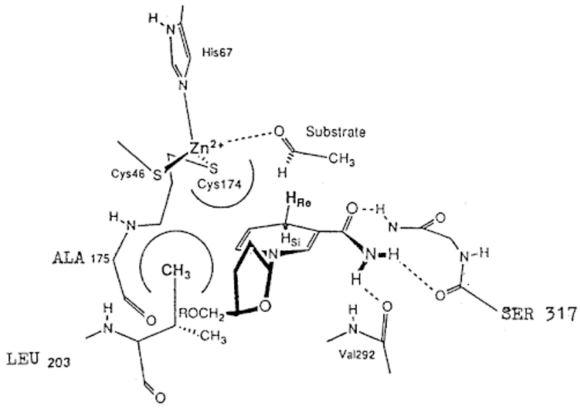


Figure 8: Proposed stereospecific mechanism of the overall reduction reaction catalyzed by the yeast ADH active site [4].

FTIR analysis of the isolated organic phase from the bioconversion experiment revealed an incomplete transformation. The infrared spectrum exhibited a dominant, sharp carbonyl stretch (C = O) at 1709.22 cm^{-1} matching the unreacted aldehyde starting material. Crucially, the characteristic broad hydroxyl (O – H) stretching band expected above 3300 cm^{-1} was entirely absent.

This lack of conversion can be explained by several experimental hurdles. First, severe emulsions formed during the liquid-liquid extraction stage with diethyl ether (Figure 9). This emulsification trapped the organic components and hindered quantitative recovery. Second, *(E)*-2-decenal exhibits cellular toxicity towards *S. cerevisiae* at millimolar concentrations, which may have deactivated the enzymatic pathways during the extended 48 h incubation period.

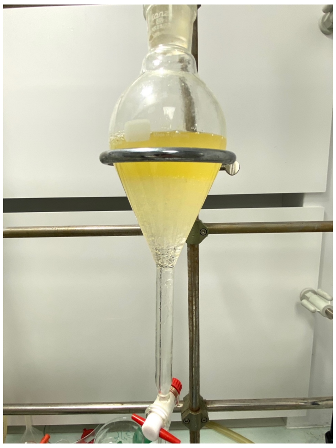


Figure 9: Visible emulsions formed between the organic and aqueous phases during liquid-liquid extraction with diethyl ether.

3.4 Inclusion Complex Characterization

The host-guest complexation of *(E)*-2-decenal within the cavity of β-CD was successfully validated. Thermodynamically, the inclusion of decenal is driven by the displacement of high-energy water molecules from the hydrophobic cavity of β-CD, which yields a positive entropy change ($\Delta S^\circ > 0$), and favorable van der Waals interactions between the guest's hydrocarbon tail and the cavity walls, driving a negative enthalpy change ($\Delta H^\circ < 0$) and overall spontaneous free energy ($\Delta G^\circ < 0$) [5].

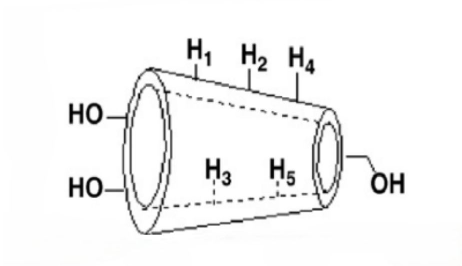


Figure 10: Structure of β-cyclodextrin illustrating the internal cavity protons (H3, H5) directly involved in host-guest interactions [7].

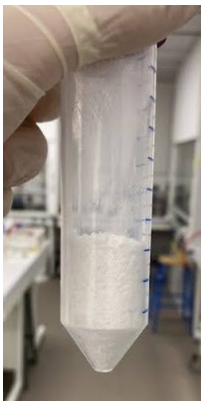


Figure 11: Visual aspect of the isolated β-cyclodextrin/*(E)*-2-decenal inclusion complex powder.

^{1}H NMR spectroscopy was used to confirm inclusion. Because the protons H3 and H5 of the complex ring are located within the interior of the cyclodextrin torus (Figure 10), any guest inclusion alters their local shielding environment, causing distinct chemical shifts ($\Delta\delta$).

Table 2: ^{1}H NMR Chemical Shifts (δ, ppm) of β-CD Protons Showing Host-Guest Intermolecular Interactions.

Proton	**β-CD + Decenal**	**Pure β-CD**	**$\Delta\delta$ (ppm)**
H1	5.047	5.051	0.004
H3	3.835	3.843	0.008
H5	3.631	3.638	0.007

As summarized in Table 2, the inner-cavity protons H3 and H5 exhibit slightly larger upfield shifts ($\Delta\delta = 0.008$ and 0.007 ppm, respectively) compared to the outer-cavity proton H1 ($\Delta\delta = 0.004$ ppm). While this systematic trend of higher $\Delta\delta$ values for the interior protons supports the hypothesis that the aliphatic chain of *(E)*-2-decenal interacts with the cyclodextrin hydrophobic core, it is important to nuance this observation. The value of these chemical shift deviations is very small and significantly lower than the typical $\Delta\delta$ values reported for robust inclusion complexes in the literature [7].

Furthermore, HS-SPME/GC-MS headspace analysis showed a measurable reduction in volatile release for the inclusion complex compared to the free aldehyde. Specifically, the normalized peak area decreased from 16.152 a.u. for the free *(E)*-2-decenal (black curve) to 15.071 a.u. upon complexation (red curve) (Figure 12). While this decrease demonstrates the trapping of the volatile aroma within the matrix and validates β-CD as a potential masking agent, the reduction remains modest. Consequently, a key objective for future improvement would be to maximize the ratio between these normalized areas, which would directly reflect a much higher and more efficient complexation rate.

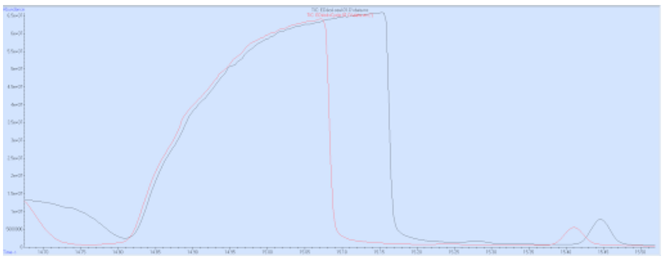


Figure 12: Comparative HS-SPME/GC-MS chromatograms of *(E)*-2-decenal in presence of β-CD (Red) and without it (Black).

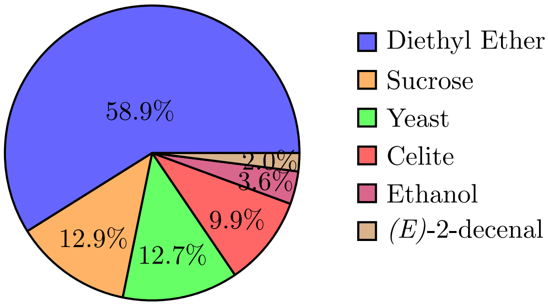
4. ENVIRONMENTAL AND ECONOMIC CONSIDERATIONS

An ecological and economic assessment was conducted to compare the two remediation pathways. Carbon footprint estimations (kg CO_2e) and cost analyses (EUR) were performed for the laboratory-scale reactions.

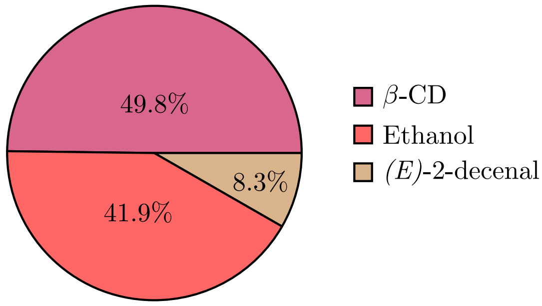
The first method (yeast bioconversion followed by solvent extraction) generated a total carbon footprint of

0.303 kg CO_2e (303 g CO_2e), largely due to the massive volume of diethyl ether required. The material cost was calculated to be EUR 8.07.

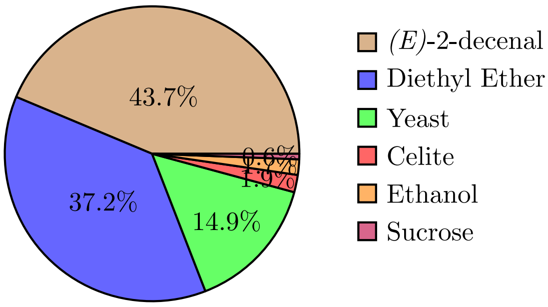
The second method (supramolecular complexation via co-precipitation) exhibited a lower carbon footprint of 0.101 kg CO_2e, alongside a reduced material cost of EUR 6.17, representing a 23.5% cost savings.



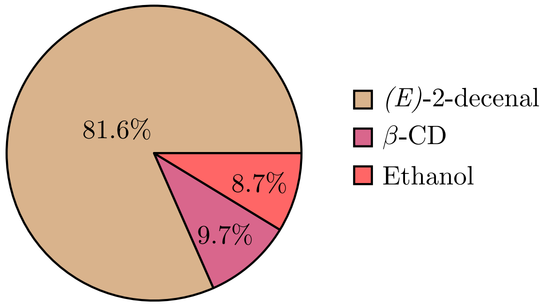
(a) Carbon Footprint (Method 1)



(b) Carbon Footprint (Method 2)



(c) Material Cost (Method 1)



(d) Material Cost (Method 2)

Figure 13: Comparative pie charts displaying the percentage distribution of the carbon footprint and material cost for the bioconversion (Method 1) and complexation (Method 2) methodologies.

This price is quite reasonable when considering a potential application such as a cyclodextrin-based mouthwash, which individuals averse to coriander could use before a meal. This environmental and economic superiority stems directly from avoiding massive volumes of volatile organic solvents and eliminating complex multi-step liquid extraction procedures.

5. CONCLUSION

This study demonstrated that while enzymatic bioconversion of the coriander allergen *(E)*-2-decenal using baker's yeast faces practical limitations, supramolecular taste-masking using β-cyclodextrin is a highly viable alternative. Formation of the host-guest inclusion complex was verified via diagnostic ^{1}H NMR chemical shifts of internal cavity protons (H3, H5) and a suppressed volatility profile in HS-SPME/GC-MS. The cyclodextrin masking strategy offers a non-toxic, inexpensive (EUR 6.17), and environmentally sustainable (0.101 kg CO_2e) approach, which could be used with further investigations for targeted mouthwash or dietary supplement solutions for coriander-sensitive individuals.

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